
Grounded AI Health Assistant & Emergency Triage

Design Document Preparation & Comprehensive Feasibility Analysis (SDG 3)

UN SDG ALIGNMENT

Goal 3: Good Health & Well-being
Target 3.8: Access to verified, safe clinical information with zero medical misinformation.

EVALUATION SCOPE

Phase 1: Design Document & Feasibility
Covers: TELOS Feasibility, SRS (FRs/NFRs), System Architecture, Schema & APIs.

CURRENT IMPLEMENTATION

Date: September 26, 2026
Status: Full Stack Deployed & Running
Tech: FastAPI · pgvector · React · Clerk

The Problem: Why General-Purpose AI Fails in Healthcare

1. The Danger of Fluent Fabrication

General LLMs are trained to always generate an answer. A confident, well-structured, completely fabricated paragraph about chest pain is more dangerous than an obvious typo because patients trust it blindly.

2. Total Lack of Provenance

Standard chatbots synthesize knowledge across unvetted internet forums (Reddit, WebMD, blogs). They cannot point to the legal document, paragraph, or clinical body behind a claim.

3. Emergency Neglect

General LLMs engage in conversational chit-chat ('I understand your chest hurts; have you tried resting?'). In acute crises, every minute lost to AI generation can be fatal.

System Philosophy: Four Architectural Guardrails

Guardrail 1: Explain, Never Diagnose

The system only explains verified medical literature. It has no capability to diagnose, prescribe medications, or suggest dosages, preventing liability and clinical drift.

Guardrail 2: 100% Auditable Provenance

Every factual statement carries an inline citation chip linking directly to a specific passage in a published document from WHO, NHS, NIH, or CDC with its legal license.

Guardrail 3: Structural Refusal (0.65 Cosine Floor)

If retrieved document similarity is below 0.65, the system refuses to answer. The LLM is never called. This structurally prevents the model from hallucinating outside verified data.

Guardrail 4: Emergency Short-Circuit (< 20 ms)

Emergency symptoms (crushing chest pain, stroke, breathing failure) are caught by rule-based pre-filters on raw input, instantly routing to 112/108 without touching the LLM.

Requirements Specification: Functional & Non-Functional (SRS)

Functional Requirements (FRs)

- FR1 (Auth & Identity): Authenticate users via Clerk JWKS RS256 JWT tokens; isolate user session histories.
- FR2 (Grounded Q&A): Retrieve relevant clinical passages from WHO/NHS/NIH/CDC and synthesize extractive responses.
- FR3 (Dynamic Citation Engine): Generate inline bracketed citation chips linking to exact source paragraphs, licenses, and URLs.
- FR4 (Emergency Triage): Scan raw text for 9 critical red-flag categories and display emergency dispatch advice (112/108).
- FR5 (Session Management): Support session creation, conversation history retrieval, and user-initiated soft-deletion.

Non-Functional Requirements (NFRs)

- NFR1 (Performance/Latency): Red-flag triage under 20 ms HTTP; warm grounded generation p50 latency < 2,500 ms.
- NFR2 (Safety & Precision): 0.0% false-hit rate on out-of-corpus queries; 100% mechanical citation validation against chunks.
- NFR3 (Availability & Fault Tolerance): Dual-provider failover (Groq -> Gemini) to guarantee 99.9% service uptime under quota exhaustion.
- NFR4 (Security & Compliance): Stateless server verification; per-user rate limiting (10 req/min); no client-side secret exposure.
- NFR5 (Maintainability): 100% automated test coverage across all critical paths (282 unit tests in < 5 seconds).

Feasibility Analysis (TELOS): Technical & Economic Viability

1. Technical Feasibility (T)

- Proven Vector Infrastructure: PostgreSQL 16 + pgvector HNSW indexing executes approximate nearest-neighbor search in under 50 ms with negligible RAM overhead.
- High-Speed LPU Inference: Groq LPU engine delivers ~1,800 ms generation times; Google Gemini Matryoshka embeddings (768d) maintain high recall.
- Seamless Streaming Architecture: FastAPI asynchronous coroutines and React fetch ReadableStream deliver real-time token rendering without WebSocket overhead.
- Empirical Proof of Concept: All 282 automated unit tests pass in 4.46 seconds, proving technical viability across all subsystems.

2. Economic / Financial Feasibility (E)

- Zero-Cost Development & Staging: \$0.00 / month operating budget using modern developer tiers:
 - Local Docker / Render Free Tier (\$0)
 - Vercel Frontend Hosting (\$0)
 - Neon Serverless Postgres + Upstash Redis (\$0)
 - Groq (8,000 TPM) & Gemini Free Tiers (\$0)
- Dramatic Cost Reduction vs Proprietary APIs:
 - Proprietary GPT-4 RAG: ~\$30.00 – \$50.00 per 1,000 queries.
 - Our Architecture: \$0.00 for development; < \$0.002 per query at commercial scale via open-weights LPUs.
- Financial Conclusion: Highly feasible for academic research, public health deployment, and resource-constrained clinics.

Feasibility Analysis (TELOS): Legal, Operational & Schedule

3. Legal & Regulatory (L)

- Not a Medical Device: Complies with CDSCO (India) and FDA SaMD guidelines by strictly framing the tool as informational. It refuses diagnostic assessment, prescribing, or drug dosage recommendations.
- Data Privacy (DPDP Act 2023 & GDPR): Collects zero Patient Health Information (PHI). Users do not upload medical scans or identity documents. User sessions support complete soft-deletion.
- Copyright Compliance: Strict attribution adhering to WHO CC BY-NC-SA 3.0, UK OGL v3.0, and US Public Domain licenses.

4. Operational Feasibility (O)

- Zero Medical Literacy Required: Designed for laypersons using natural, conversational language on mobile and desktop.
- Human-in-the-Loop Paradigm: Does not replace doctors; works as an informational guide and signposts users to 112/108 or hospital clinics.
- Operational Resilience: Rate limiting prevents abuse; automatic provider fallback prevents downtime.

5. Schedule Feasibility (S)

- Phase 1 (26 Sep 2026 - Today): Design Document Preparation, Feasibility Study, and Architecture Specification [COMPLETED].
- Phase 2 (31 Oct 2026): Prototype Validation, Clinical Retrieval Benchmark Suite, and Fallback Stress-Testing [ON TRACK].
- Phase 3 (01 Dec 2026): Final Production Deployment, User Study & Evaluation Defense [SCHEDULED].

System Architecture: End-to-End Pipeline Data Flow

Step 1: Auth & Rate Limit

Clerk JWT (RS256 JWS)
Redis Lua Sliding Window
10 req/min per user
(Fails open)

Step 2: Red-Flag Triage

In-Process Regex (< 1 ms)
Emergency Match -> Fixed 112/108
(No Embedding, No LLM)

Step 3: Vector Search

Gemini 768d Embeddings
Postgres 16 + pgvector HNSW
Cosine similarity floor ≥ 0.65

Step 4: Dual LLM Fallback

Primary: Groq (Llama/GPT-OSS)
Fallback: Gemini 2.5 Flash
Zero broken streams on error

Step 5: SSE Streaming UI

React 18 + Vite + Tailwind
Server-Sent Events streaming
Interactive Citation Chips

Database Design: Relational Schema & pgvector HNSW Index

documents (Corpus Metadata)

- id: UUID (Primary Key)
- title: Text (Published article title)
- source_url: Text (Original reference link)
- source_org: Text (WHO | NHS | NIH | CDC)
- license: Text (OGL-v3 | CC BY-NC-SA | Public)
- content_hash: String(64) UNIQUE (SHA-256 for idempotent re-ingestion)
- ingested_at: DateTime(timezone=True)

chunks (Vector Store)

- id: UUID (Primary Key)
- document_id: UUID (FK -> documents.id CASCADE)
- content: Text (Semantic chunk passage)
- embedding: Vector(768) (Gemini 768-dim vector)
- chunk_index: Integer (0-based chunk order)
- token_count: Integer (~300 to ~650 tokens)
- ★ INDEX: HNSW on embedding (vector_cosine_ops, m=16, ef_construction=64)

sessions (User Chat Context)

- id: UUID (Primary Key)
- clerk_user_id: Text (Indexed, user scope)
- title: Text (Auto-generated from query)
- deleted_at: DateTime (Nullable, soft delete)
- created_at: DateTime(timezone=True)

messages (Audit Trail & Tracking)

- id: UUID (Primary Key)
- session_id: UUID (FK -> sessions.id CASCADE)
- role: Text ('user' | 'assistant')
- content: Text (Message body or response)
- retrieved_chunk_ids: UUID[] (Audit trail)
- llm_provider: Text ('groq' | 'gemini')
- latency_ms: Integer | was_red_flag: Boolean

API Architecture: REST & Server-Sent Events (SSE) Endpoints

CORE API SPECIFICATION

- **POST /api/chat/stream [Bearer <JWT>]**
Payload: {session_id, content} -> SSE stream with event types: token, citation, escalation, done
- **GET /api/sessions [Bearer <JWT>]**
Payload: None -> Returns 200 OK with list of active user chat sessions [SessionOut]
- **GET /api/sessions/{id}/messages [Bearer <JWT>]**
Payload: session_id -> Returns 200 OK with full conversation history & citation IDs [MessageOut]
- **DELETE /api/sessions/{id} [Bearer <JWT>]**
Payload: session_id -> Soft-deletes session (sets deleted_at); returns 204 No Content
- **GET /api/me [Bearer <JWT>]**
Payload: None -> Returns 200 OK with authenticated Clerk user ID & email (401 if invalid)
- **GET /health [Public]**
Payload: None -> Returns 200 OK: {status, database: {reachable, pgvector}, redis: {reachable}}

Security & Middleware Layer

- RequestContextMiddleware: Injects unique X-Request-ID across all logs.
- Clerk JWKS Authenticator: Validates RS256 JWT tokens cryptographically.
- Redis Sliding Window: Enforces 10 req/min rate limit per user.

SSE Streaming Protocol (ReadableStream)

- Uses fetch + ReadableStream instead of native EventSource to support Authorization headers.
- Dispatches real-time structured JSON chunks to React frontend.
- Automatically cleans up DB sessions and logs response latency.

The Medical Reference Corpus: Quality Over Unvetted Breadth

Curated & License-Checked Repository

- 42+ Vetted Source Documents: Covering infectious diseases (dengue, malaria, TB), chronic conditions (diabetes, hypertension, asthma), everyday symptoms, and prevention.
- Strict Licensing Compliance: Every source explicitly mapped in corpus_manifest.json:
 - WHO: CC BY-NC-SA 3.0 IGO
 - UK NHS: Open Government Licence (OGL v3.0)
 - US NIH / CDC: US Federal Public Domain
- Deterministic Extraction: Handled via PyMuPDF and Trafilatura, stripping boilerplate, headers, and footers.
- Semantic Chunking: 153 chunks bounded between ~300 to ~650 tokens to ensure clinical coherence.

High Priority for Indian Healthcare

- High-Relevance Infectious Disease Focus: Extensive coverage for Dengue, Malaria, Tuberculosis, Typhoid, and Diarrhoeal diseases.
- Everyday Family Symptoms: High fever in adults & children, cough, dehydration, vomiting, paediatric skin rashes.
- Deliberate Frozen Corpus: Frozen snapshot prevents silent data corruption or unvetted web drift. If a condition isn't verified, it safely refuses.
- Auditable Citations: Every answer references the exact source document title, URL, and medical authority.

Safety Engineering: Deterministic Emergency Short-Circuit

PERFORMANCE ASYMMETRY: EMERGENCY TRIAGE vs GENERATION

Red-Flag Matcher: < 1 ms in-process (~20 ms HTTP) vs LLM Generation: ~2,512 ms

Short-circuits immediately before embedding, retrieval, and LLM generation. Eliminates latency and non-deterministic risks.

9 Critical Emergency Categories

1. Cardiac (crushing chest pain, arm radiation)
2. Breathing Distress (choking, severe asthma)
3. Stroke Signs (BE-FAST: facial droop, arm weakness)
4. Uncontrolled Severe Bleeding
5. Acute Head Injury with altered mental state
6. Loss of Consciousness / Unresponsiveness
7. Anaphylaxis (throat swelling, severe allergy)
8. Seizures / Convulsions
9. Mental Health Crisis / Self-Harm

Emergency Helplines & Asymmetric Cost

Integrated Indian Emergency Services:

- 112: All-in-One National Emergency Helpline
- 108: Emergency Ambulance Response
- 14416: Tele-MANAS (Mental Health Helpline)

The 'Cost Asymmetry' Rule:

If a user says 'I know it's just acid reflux, but my chest feels tight', the system still escalates. An unnecessary emergency checkup is an annoyance; a missed heart attack is fatal.

High Availability & Operational Resilience

Dual-Provider Automatic LLM Fallback

- Primary Provider: Groq (Llama / GPT-OSS 120B) for ultra-fast token streaming (~1,800 ms median latency).
- Automatic Fallback: Gemini 2.5 Flash triggers automatically if Groq hits rate limits (TPM), connection timeouts (8s limit), or auth outages.
- Clean Stream Recovery: If the primary fails mid-generation, the client cleanly discards the partial buffer and resumes from the fallback. No broken hybrid text.
- Empirically Verified: We validated that invalidating Groq's key immediately triggers Gemini with zero client crashes.

Redis Sliding-Window Rate Limiting

- Per-User Sliding Window: 10 requests per minute enforced per authenticated Clerk User ID.
- Atomic Lua Script: Trims old timestamps, counts active requests, records timestamp, and sets TTL in a single atomic Redis round-trip.
- Eliminates Race Conditions: Traditional read-then-write approaches permit concurrent limit bypass. Atomic execution prevents this.
- Fail-Open Resiliency: If Redis is unavailable, rate limiting is bypassed. A rate-limiter failure never takes down critical health access.

Current Implementation Status: Production Ready

Backend & Database

FastAPI (Python 3.11, async)
SQLAlchemy 2.0 + asyncpg
PostgreSQL 16 + pgvector (HNSW)
Redis 7 Ephemeral Cache
Alembic Database Migrations

Frontend & UX

React 18 + TypeScript + Vite
Tailwind CSS modern styling
Server-Sent Events (SSE) Reader
Interactive Citation Inspector
Emergency Visual Escalation

Security & Auth

Clerk Authentication
JWKS Cryptographic Verification
RS256 JWT validation on all API calls
No browser-side credential trust
Strict CORS & rate limit headers

RIGOROUS AUTOMATED TEST SUITE: 282 UNIT TESTS (100% PASSING)

Comprehensive test coverage executed in 4.46 seconds:

- 98 tests for red-flag emergency detection & phrasing variants
- 38 tests for vector retriever & floor filtering
- 31 tests for document extraction & chunking bounds
- 21 tests for authentication & rate limiting

Empirical Benchmark Results: Grounding & Latency

Retrieval & Grounding Precision

- In-Corpus Retrieval Hit Rate: 90% (27/30 in-corpus medical queries retrieved relevant passages).
- Out-of-Corpus False Hit Rate: 0.0% (0/8 non-medical or off-topic queries cleared the 0.65 threshold).
- Top Similarity Separation: In-corpus median: 0.727 vs Out-of-corpus max: 0.555 (Wide 0.172 margin).
- Citation Validity: 100% (129 of 129 generated citations mechanically verified to match retrieved source text).
- Mean Citations: 4.8 verified sources cited per response.

System Latency Profile

- Red-Flag Escalation: ~0.1 ms in-process matcher (HTTP round-trip ~20 ms).
- Query Embedding (Gemini): Median 714 ms (p95: 950 ms).
- Vector Retrieval (pgvector HNSW): Median 46 ms (p95: 48 ms).
- Time to First Token (TTFT): Median 1,129 ms.
- Total End-to-End Generation: Median 2,512 ms (Groq p50: 1,843 ms, Gemini fallback p50: 3,605 ms).
- Safety Short-Circuit: 100% of red-flag queries prevented from reaching the LLM.

Live Demonstration: Four Key Evaluation Beats

Beat 1: Grounded Answer

Query: 'What causes iron deficiency anaemia?'

Shows: Real-time SSE streaming, inline citation chips, and source inspector panel displaying NHS/WHO origin.

Beat 2: Grounding Proof

Query: 'What is the capital of France?'

Shows: Immediate refusal. Proves the model is structurally blocked without verified medical context.

Beat 3: Emergency Short-Circuit

Query: 'Crushing chest pain radiating to arm'

Shows: Instant red emergency banner (~20 ms). Directs to 112/108 without waiting for LLM tokens.

Beat 4: Cost Asymmetry

Query: 'I think it's acid reflux but chest hurts'

Shows: Still escalates. Proves safety priority: false alarms are minor; missed heart attacks are fatal.

Defense: Why Use This Over Frontier AI Models (ChatGPT/Gemini)?

Frontier AI Models (ChatGPT, Claude)

- Unconstrained Intelligence: Trained to be helpful and conversational. They will speculate on symptoms and suggest unverified home remedies.
- Opaque Sourcing: Synthesize text from internet scrapings (Reddit, forums, blogs). Zero legal traceability or document-level citations.
- Dangerous Latency in Crises: Take 2-5 seconds over HTTP to chat conversationally when a user is experiencing cardiac arrest or stroke.
- Prompt Drift & Jailbreaks: System prompts ('Act like a doctor') can be bypassed or ignored by creative user queries.

Our Grounded Assistant Architecture

- Enforced Negative Constraints: Architecturally incapable of hallucinating without verified medical context (0.65 similarity floor).
- 100% Chain-of-Custody: Every claim links to an exact legal document chunk from WHO, NHS, NIH, or CDC.
- Deterministic Emergency Speed: Rule-based regex evaluates raw message in < 1 ms (~20 ms HTTP) and routes to 112/108 immediately.
- Zero Diagnosis Liability: Strictly explains literature; cannot prescribe medication or suggest dosages by design.

Engineering Boundaries & Future Roadmap

Current Engineering Boundaries

- Curated Frozen Corpus: Limited to 42-50 vetted documents. Questions outside this scope (e.g. rare diseases, COVID-19) are refused.
- Informational Only: Built strictly as an informational tool. Cannot replace physical medical exams, diagnostics, or clinical triage.
- English Language: Currently trained and indexed for English medical literature.
- Manual Page Extraction for Bot-Protected Sites: 8 CDC articles require manual saving due to strict anti-scraping Cloudflare blocks.

Future Technical Roadmap

- Automated LLM-as-a-Judge Verification: Adding real-time claim-to-passage entailment verification (NLI) before streaming.
- Indian Regional Language Support: Expanding ingestion and queries to Hindi, Tamil, Telugu, and Bengali.
- National Health Authority Ingestion: Expanding corpus to include official publications from ICMR (Indian Council of Medical Research) and AIIMS.
- Offline Edge Deployment: Exploring quantized on-device embeddings and small language models for remote rural health clinics.

Summary: An Assistant You Can Audit and Trust

Safety-critical AI requires architectural constraints, deterministic triage, and legal provenance.

100%

Grounded Citation Validity (Verified against retrieved chunks)

< 20 ms

Emergency Triage Short-Circuit to 112 / 108

282

Passing Automated Unit Tests covering every safety layer